

Introduction to

ITI - Intake 46

^ SOFTWARE ENGINEERING

Beshoy adel

Evaluation Criteria

- Duration: 4 days (**2 lectures**).
- **30%** Attendance, Activities and Class Participation.
- **70%** Presentation&Project .

Topics Covered

- **Professional software development**

What is meant by software engineering and related topics.

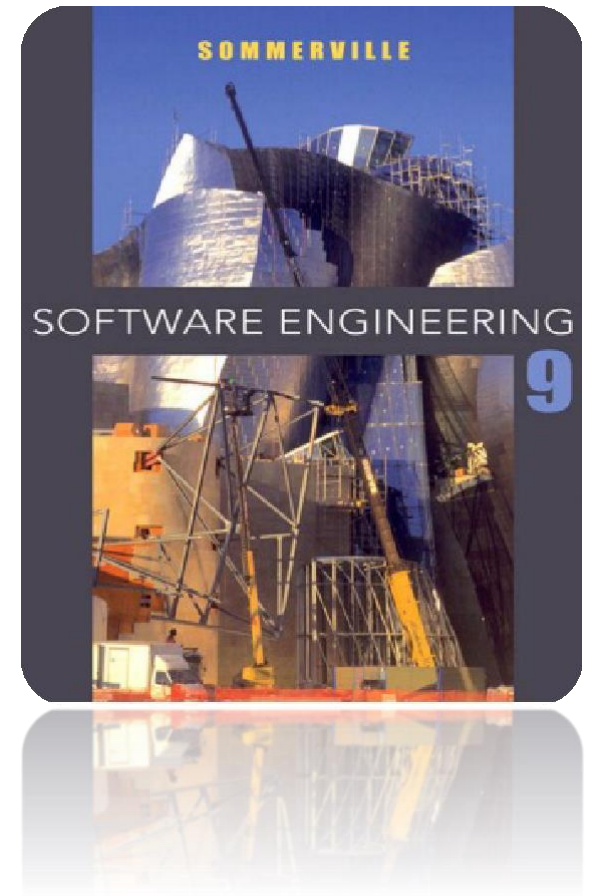
- **Software engineering ethics**

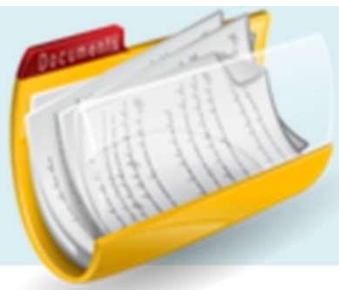
A brief introduction to ethical issues that affect software engineering.

- **Case studies**

An introduction to three examples that are used in later chapters in the book.

(SWE Ian Sommerville 9 edition)





Agenda

► DAY 1

- Program & Software
- Application types
- Good Software Attributes
- Software impact
- software Engineering
 - SWE Roles
 - SWE Skills
- SDLC





Agenda

► DAY 2

- Software Testing
 - Error ,Bugs , Defect , Fault & failure
 - Testing stages and Types
 - Quality Assurance (QA)
 - Quality Control (QC)
- Software Models
 - Waterfall Model
 - Incremental Model
 - Spiral Model
 - Prototype
- Software engineering Ethics
- DAY 3 UML
- DAY 4 Presentation



“Programming today is a race between software engineers striving to build bigger and better idiot-proof programs, and the Universe trying to produce bigger and better idiots. So far, the Universe is winning.”

—Rick Cook

Programming is NOT enough !



What is

software ??

Program

- **A program :** is a set of instructions executed by computer to perform a specific task.
- **Application:** A program or group of programs that is designed for the end user. Application software (an application) is a set of computer programs designed to allow users to perform a group of coordinated functions, tasks, or activities.

What is **software**?



Computer Programs

+



associated

^ Documentation



Documentation Types

- ▶ **User Documentation**
- ▶ **Technical Documentation**
- ▶ **Marketing Documentation**





Software types

- ▶ **Generic**
- ▶ **Customized**





Application types

Activity Time





Application types

- ✓ **Stand- Alone**
- ✓ **Interactive transaction-based**
- ✓ **Embedded Control Systems**
- ✓ **Entertainment Systems**
- ✓ **Systems for modeling and simulation**
- ✓ **Data Collection Systems**
- ✓ **System of systems**



Application types

- **Stand-alone applications**
 - These are application systems that run on a local computer, such as a PC. They include all necessary functionality and do not need to be connected to a network.
- **Interactive transaction-based applications**
 - Applications that execute on a remote computer and are accessed by users from their own PCs or terminals. These include web applications such as e-commerce applications.
- **Embedded control systems**
 - These are software control systems that control and manage hardware devices. Numerically, there are probably more embedded systems than any other type of system.



Application types

- **Entertainment systems**
 - These are systems that are primarily for personal use and which are intended to entertain the user.
- **Systems for modeling and simulation**
 - These are systems that are developed by scientists and engineers to model physical processes or situations, which include many, separate, interacting objects.



Application types

- **Data collection systems**
 - These are systems that collect data from their environment using a set of sensors and send that data to other systems for processing.
- **Systems of systems**
 - These are systems that are composed of a number of other software systems.



Attributes of Good Software





Attributes of Good Software

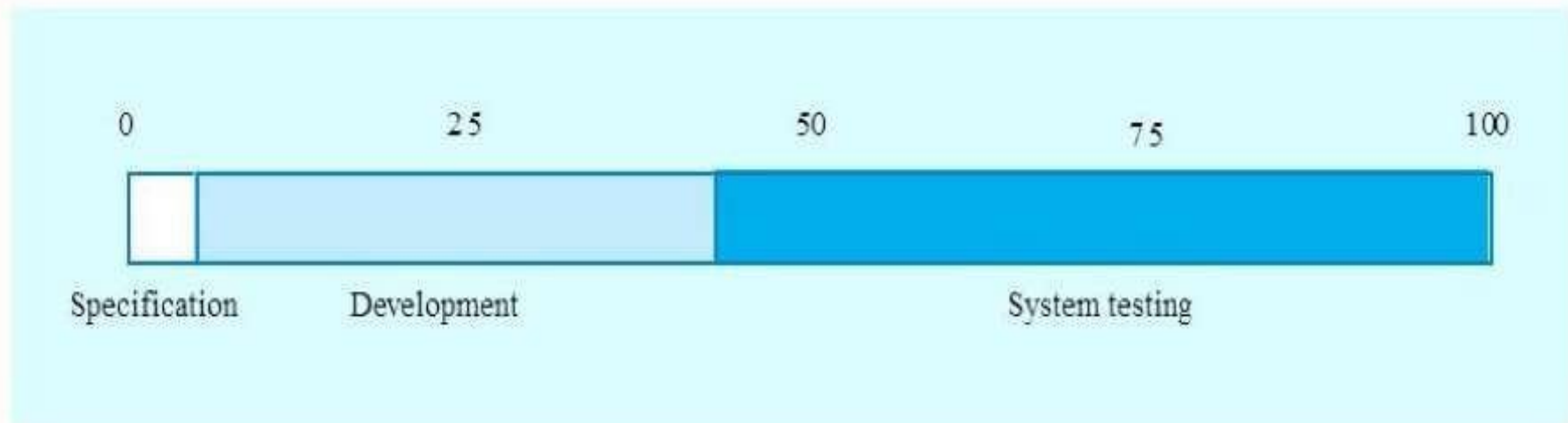




Software Cost

60% Development

40 % testing

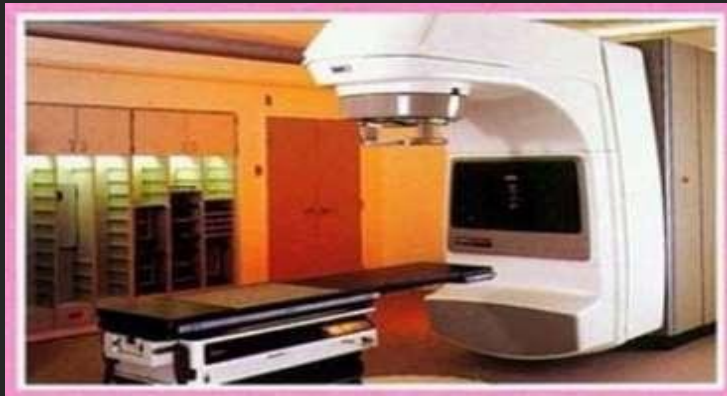




Importante of software

Software can have a huge impact in any aspect of society.

Therac-25



PATIENT NAME: John		BEAM TYPE: E	ENERGY (KeV):	10
TREATMENT MODE: FIX				
		ACTUAL	PREScribed	
UNIT RATE/MINUTE	0.000000	0.000000		
MONITOR UNITS	200.000000	200.000000		
TIME (MIN)	0.270000	0.270000		
GANTRY ROTATION (DEG)	0.000000	0.000000	VERIFIED	
COLLIMATOR ROTATION (DEG)	359.200000	359.200000	VERIFIED	
COLLIMATOR X (CM)	14.200000	14.200000	VERIFIED	
COLLIMATOR Y (CM)	27.200000	27.200000	VERIFIED	
WEDGE NUMBER	1.000000	1.000000	VERIFIED	
ACCESSORY NUMBER	0.000000	0.000000	VERIFIED	
DATE: 2012-04-16	SYSTEM: BEAM READY	OP.MODE: TREAT	AUTO	
TIME: 11:48:58	TREAT: TREAT PAUSE	X-RAY	173777	
OPR ID: 033-ufsp	REASON: OPERATOR	COMMAND: █		

Between 1985 and 1987, at least six patients died as a consequence of severe overdoses of radiation delivered.

The cause was a fault in the control software.

Ariane 5 Flight 501 “1996”



Cause: design errors in the software

[Video](#)

Chaos Report 'Standish Group'



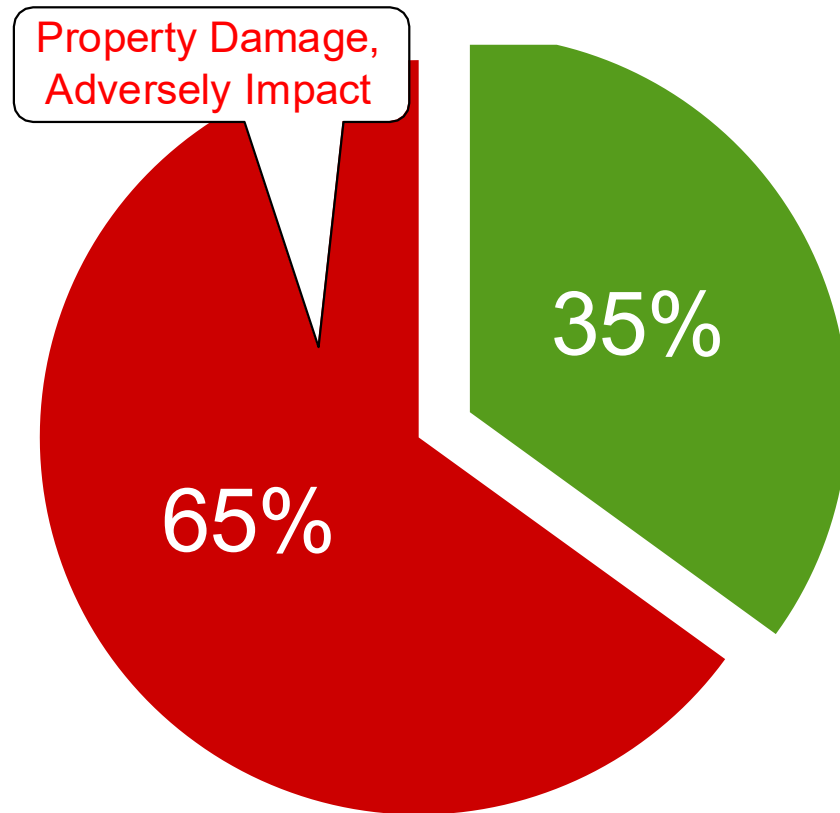
Chaos Report

MODERN RESOLUTION FOR ALL PROJECTS

	2011	2012	2013	2014	2015
SUCCESSFUL	29%	27%	31%	28%	29%
CHALLENGED	49%	56%	50%	55%	52%
FAILED	22%	17%	19%	17%	19%

The Modern Resolution (OnTime, OnBudget, with a satisfactory result) of all software projects from FY2011 - 2015 within the new CHADS database. Please note that for the rest of this report CHADS Resolution will refer to the Modern Resolution definition not the Traditional Resolution definition.

Most projects **fail** than **succeed**



■ 65% Completed Late, Over Budget, Does not meet the requirements, Feature Missing, Not Easy to Use, Canceled

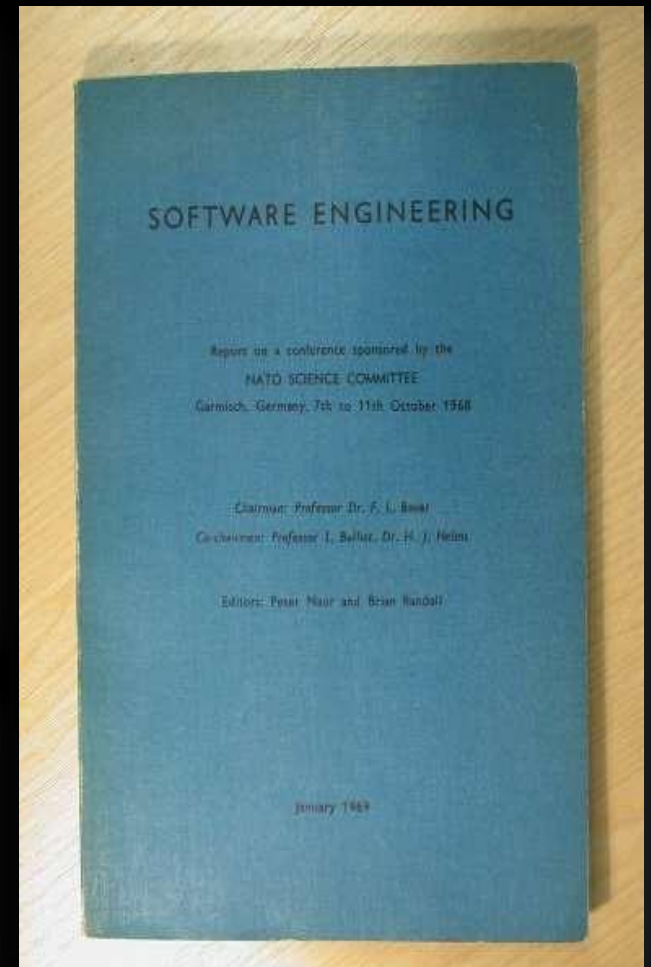
■ 35% Successful



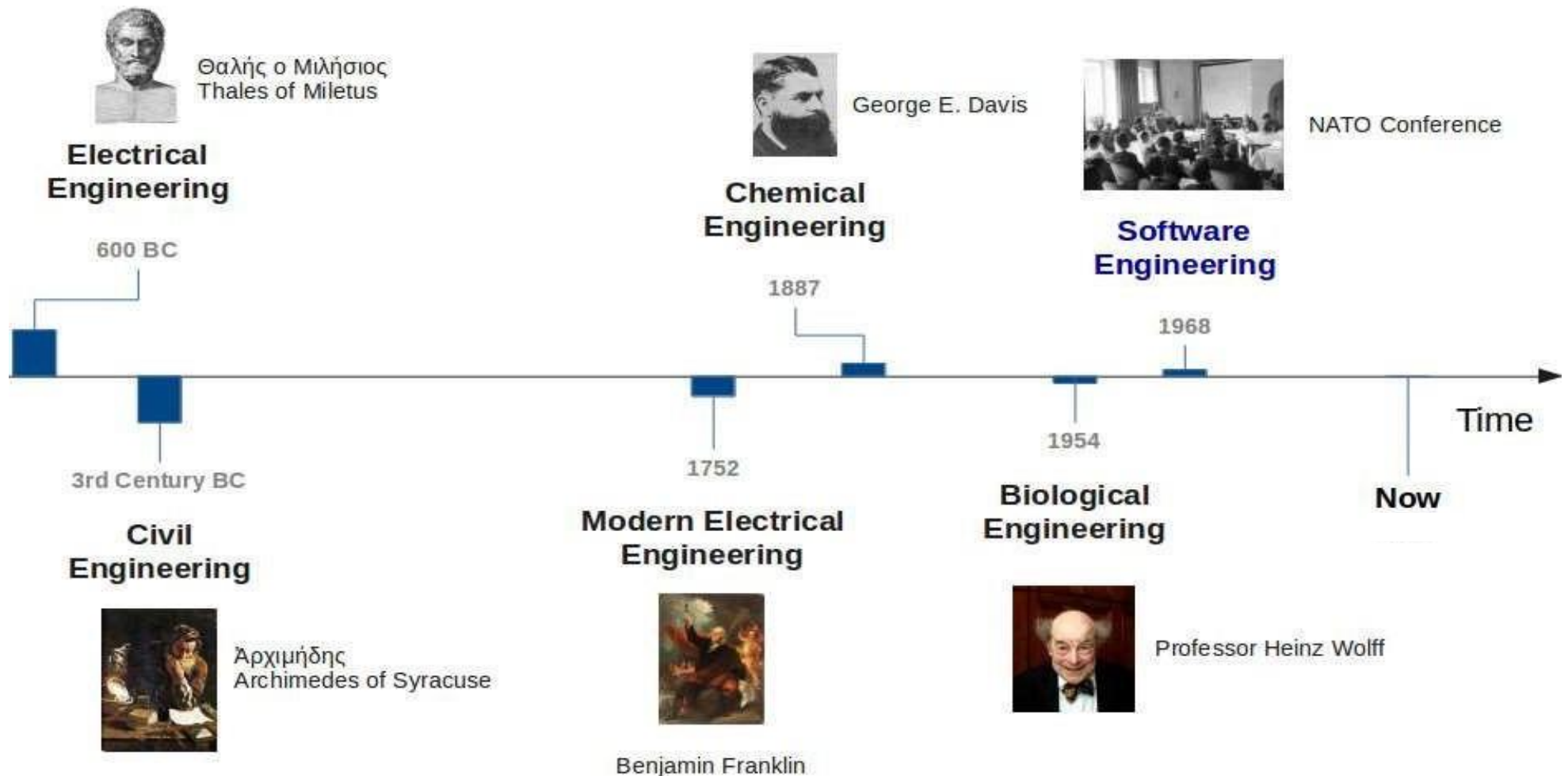
Projects **fail !! WHY?**



The origins of software engineering: NATO conference in 1968



the origin of software engineering and other fields



The role of software engineering is to capture the customer's business needs and specify the “blueprints” for the system so that programmers can implement it



What is software engineering?

Software engineering:

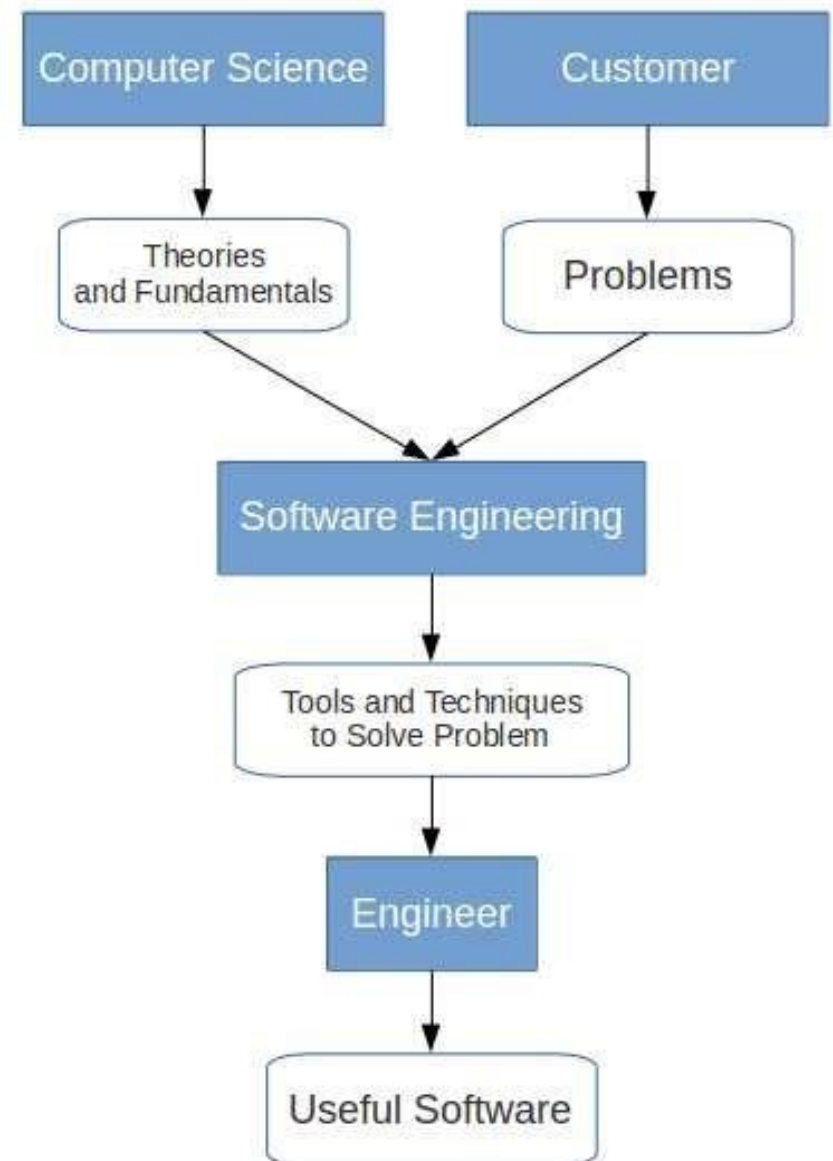
is an engineering discipline that is concerned with all aspect of software production.

Most effective way
to produce high-quality
software



Software engineering vs computer science

- ❖ Computer science focuses on theory and fundamentals.
- ❖ Software engineering is concerned with the practicalities of developing and delivering useful software.



The Team



12

Daily
Software
Engineering
Jobs/ Roles

1 Requirement Engineer

2 Systems Analyst

3 Designer

4 UI/UX developer

5 Software Programmer

6 Software Systems Administrator

7 Software Database Administrator

8 Software Security Engineer

9 Software Tester

10 Software Project Manager

11 Software Configuration Engineer

12 IT Help Desk

Position Levels

- Trainee / fresh
- Junior
- Mid level
- Senior
- Team Leader
- Project Manager



Software Engineering

Skills



Communication



**“There are only 10 types of people in this world.
Those who know binary, and those who don’t.”**

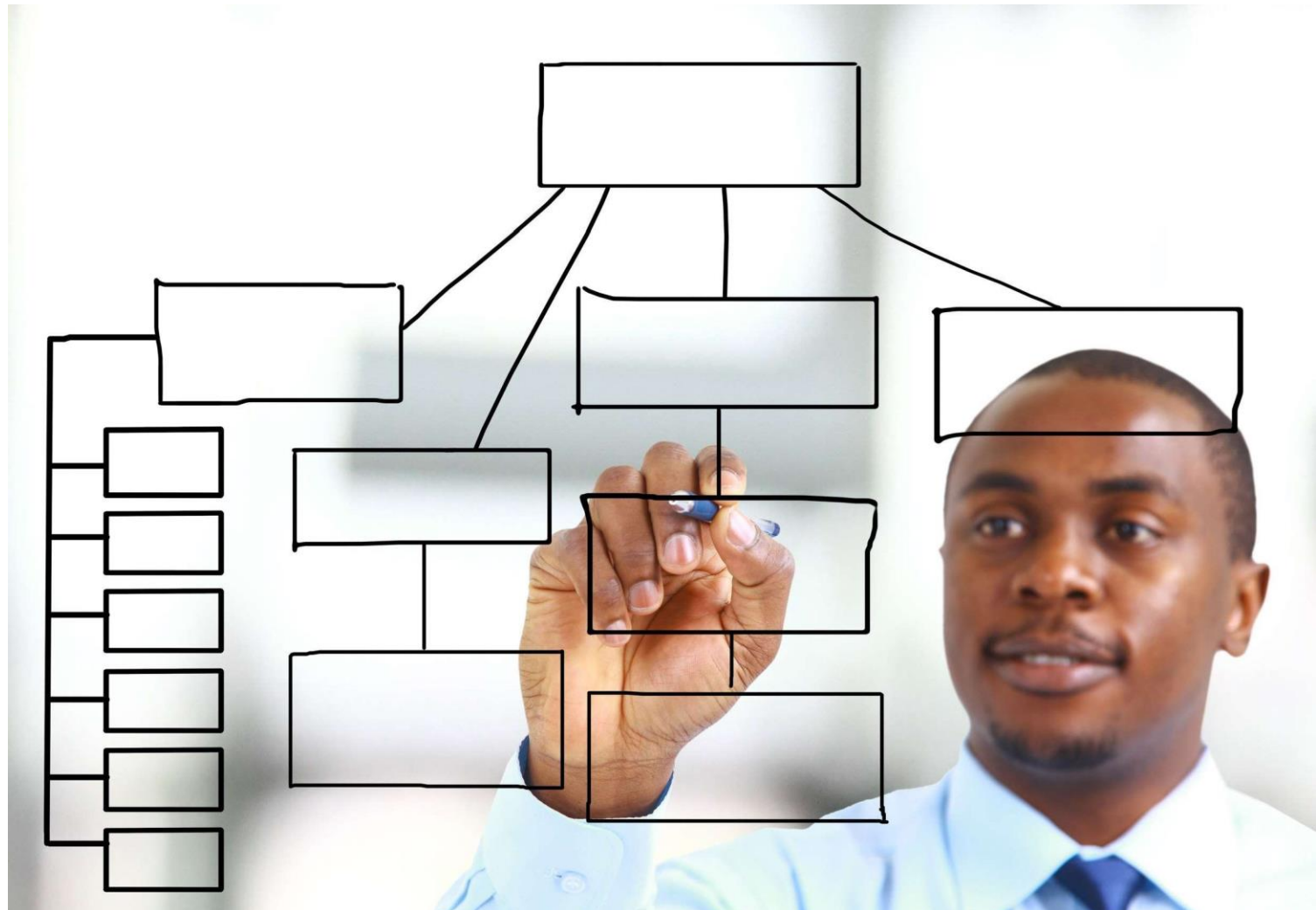


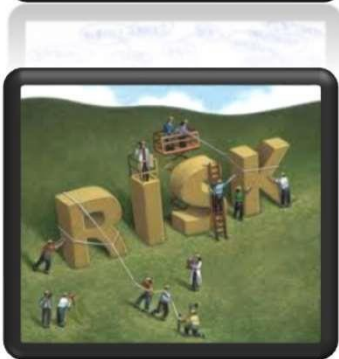
Feedback



Motivation

Analytical





Risk Assessment

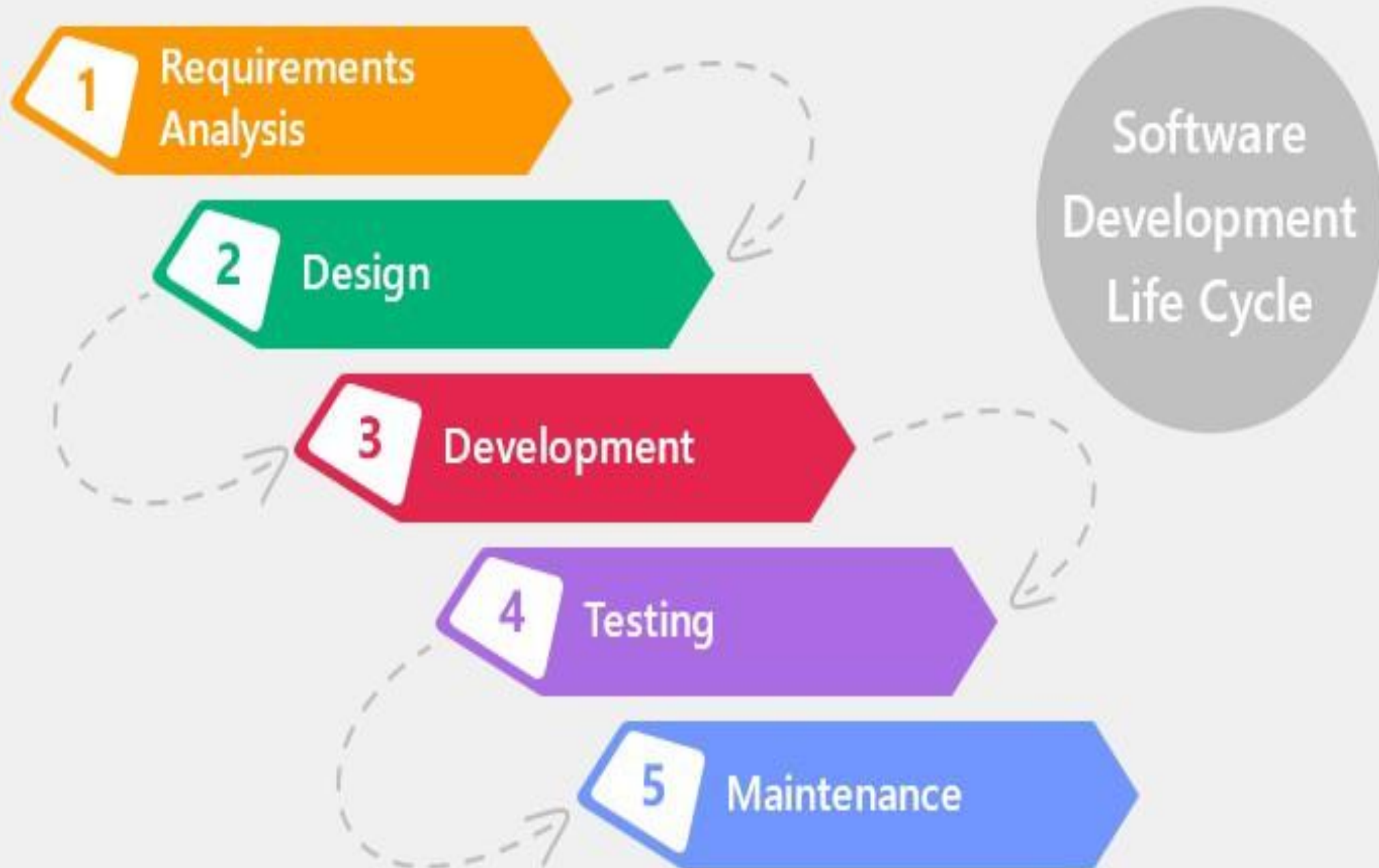
Estimation

Software Process

- A software process (also known as software methodology) is a set of related **activities** that leads to the **production** of the software. These activities may involve the development of the software from the scratch or modifying an existing system.



Stages for software development



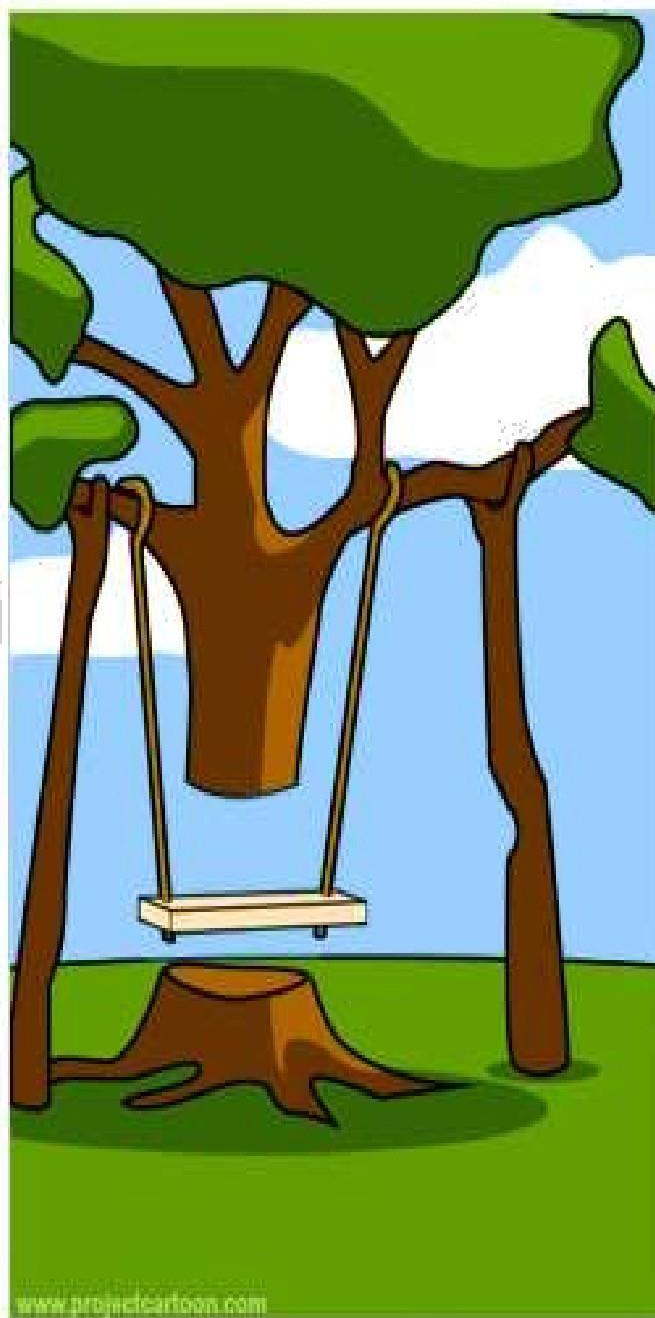
Requirement

- descriptions of what the system should do.
- These requirements reflect the needs of customers for a system that serves a certain purpose , it provides the constraints on its operation.





How the customer explained
it



How the team designed it



What the customer really
needed

Problems of requirements elicitation

- Stakeholders don't know what they really want.
- Stakeholders express requirements in their own terms.
- Different stakeholders may have conflicting requirements.
- The requirements change during the analysis process. New stakeholders may emerge and the business environment change.



Requirements Engineering

- The process of finding out, analyzing, documenting and checking these services and constraints



Interviewing

interviews with stakeholders are part of most Reprocesses.

- **Types of interview**

- ✓ **Closed interviews** based on pre-determined list of questions
- ✓ **Open interviews** where various issues are explored with stakeholders.

- **Effective interviewing**

- ✓ Be open-minded, avoid pre-conceived ideas about the requirements and are willing to listen to stakeholders.
- ✓ Prompt the interviewee to get discussions going using a springboard question, a requirements proposal or by working together on a prototype system.



Interviews in practice

- Normally a mix of closed and open-ended interviewing.
- Interviews are good for getting an overall understanding of what stakeholders do and how they might interact with the system.
- Interviews are not good for understanding domain requirements



Scenarios

- Scenarios are real-life examples of how a system can be used.
- They should include
 - A description of the starting situation;
 - A description of the normal flow of events;
 - A description of what can go wrong;
 - Information about other concurrent activities;
 - A description of the state when the scenario finishes.



Ethnography

- A social scientist spends a considerable time observing and analysing how people actually work.
- People do not have to explain or articulate their work.
- Social and organisational factors of importance may be observed.
- Ethnographic studies have shown that work is usually richer and more complex than suggested by simple system models.



Levels of description Requirements

- **User requirements** : are statements, in a natural language, of what services the system is expected to provide to system users.

Ex: “The System shall generate monthly management reports showing the Profit for each product prescribed by each Branch during that month.”



Levels of description Requirements

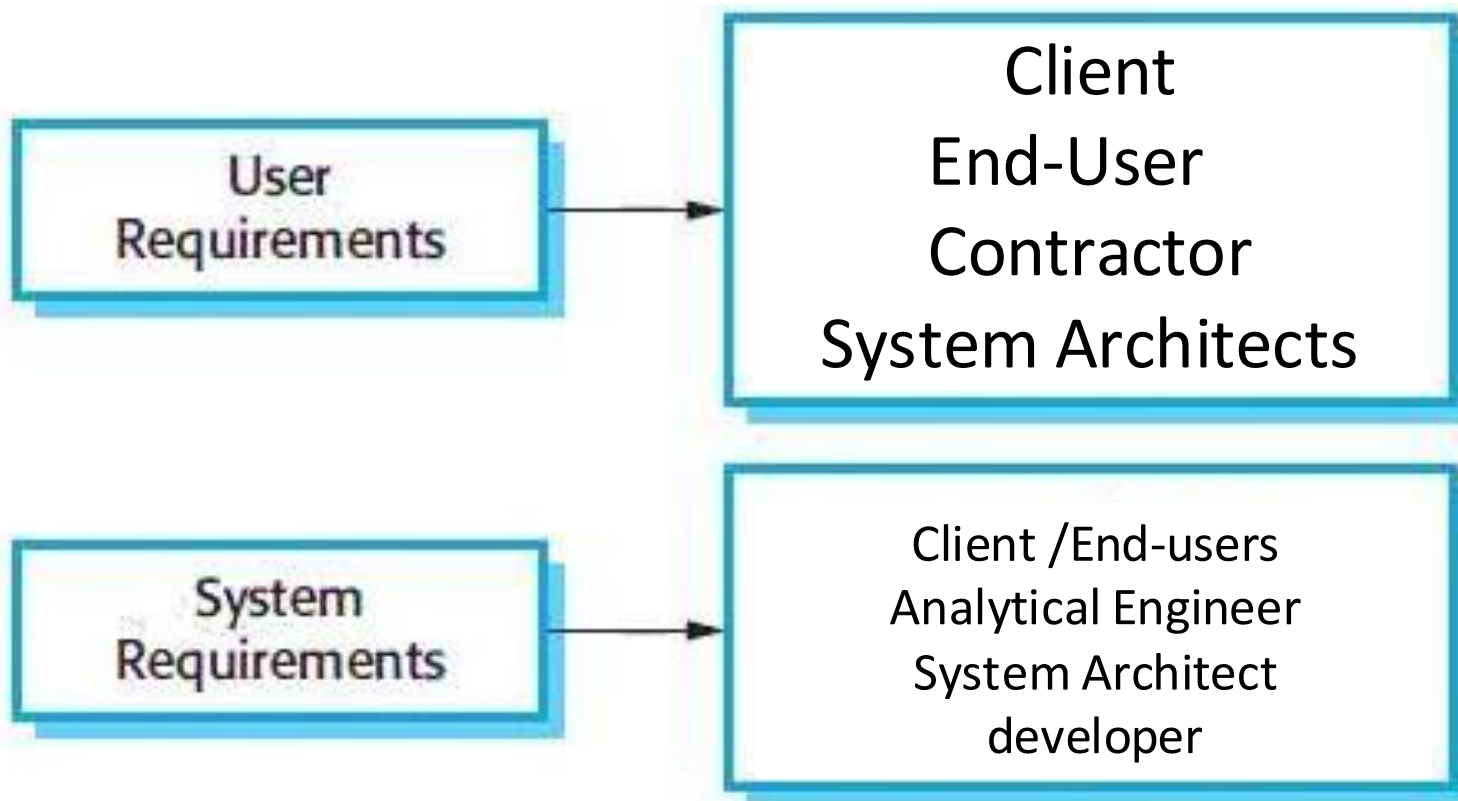
- **System requirements (SRS)** : are more detailed descriptions of the software system's functions, services, and operational constraints.
- **should define exactly what is to be implemented.**
- It may be part of the contract between the system buyer and the software developers.



System requirements

- “1.1 On the last working day of each month, a summary of the products , their profit , and the branch shall be generated.
- 1.2 The system shall automatically generate the report for printing after 7.30 on the last working day of the month.
- 3. A report shall be created for each branch and shall list the individual Product names, the total number of this product , the number of sold, and the total Profit of this product .
- 4. If Product are available in Different sizes(e.g., 10 mg, 20 mg) separate reports shall be created for each Size unit.
- 5. Access to all profit reports shall be restricted to authorized users listed on a management access control list.”

Who Read it !!?



System requirements classification

- ***Functional requirements*** : These are statements of services the system should provide, how the system should react to particular inputs, and how the system should behave in particular situations.
- ***Non-functional requirements*** : These are constraints on the services or functions offered by the system. They include timing constraints, constraints on the development process

Types of non-functional requirement

- Access
- Usability
- Cost
- Capacity
- Documentation
- recovery
- Efficiency
- Backup
- licensing
- Performance
- Platform compatibility
- Privacy
- Quality
- Response time
- Security
- Support
- Test
- Speed

software requirements document

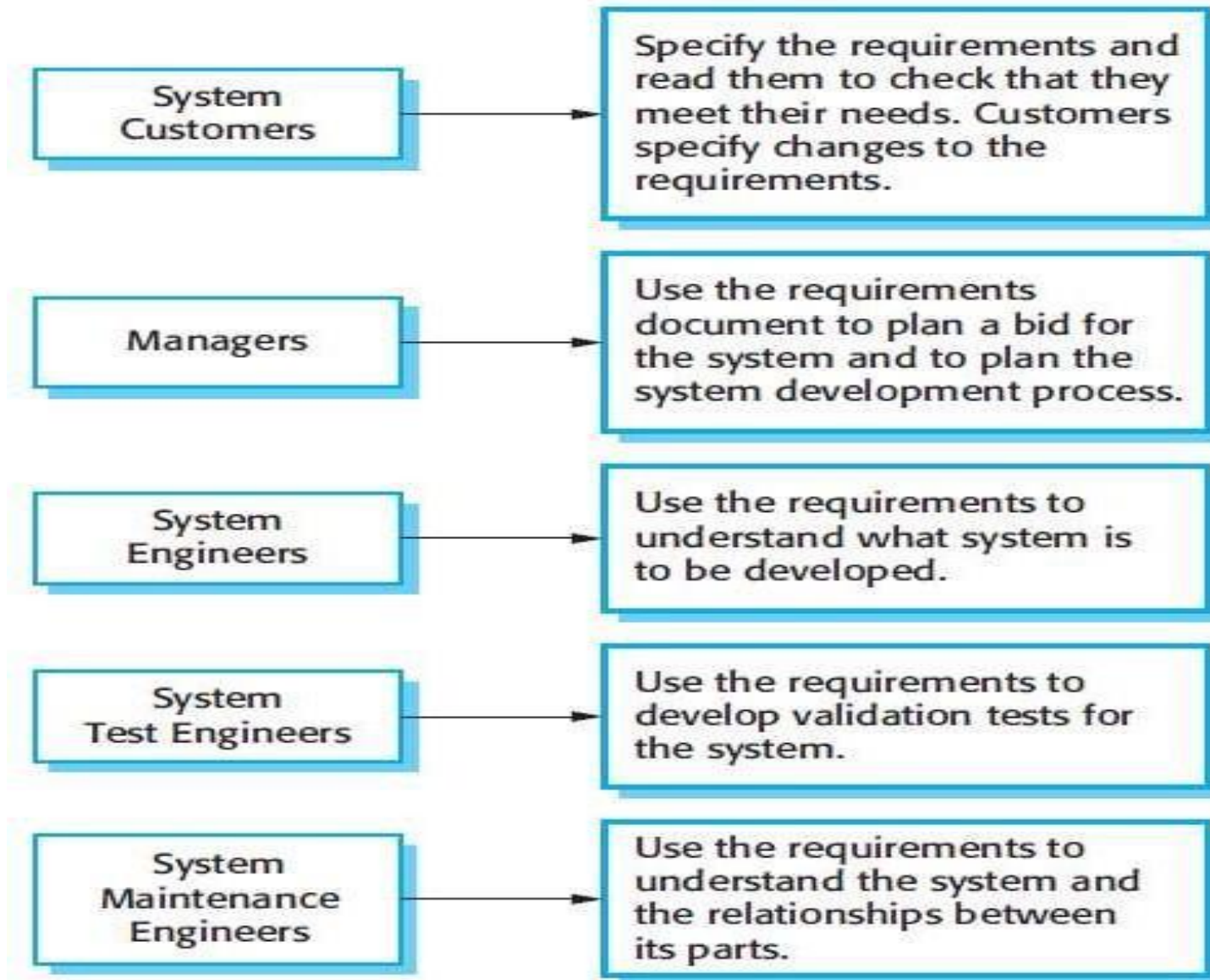
- sometimes called the software requirements specification or SRS
- is an official statement of what the system developers should implement.
- It should include both the user requirements for a system and a detailed specification of the system requirements.



SRS

**Software Requirements
Specification**

Users of a requirements document



Requirements validation

- Concerned with demonstrating that the requirements define the system that the customer really wants.
- Requirements error costs are high so validation is very important
 - Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error.





How the customer explained
it



How the team designed it



What the customer really
needed

Requirements checking

- Does the system provide the functions which best support the customer's needs?
- Are there any requirements conflicts?
- Are all functions required by the customer included?
- Can the requirements be implemented given available budget and technology?

2. Design

The image shows a vertical stack of three user interface designs, each within its own frame. The top frame, titled 'TYPICAL APPLE PRODUCT...', contains a large square button with a circle in the center and the word 'TOUCH' inside. The middle frame, titled 'A GOOGLE PRODUCT...', contains a search bar with a 'FIND' button to its right. The bottom frame, titled 'YOUR COMPANY'S APP...', contains a complex form with many fields, checkboxes, and buttons. The form includes fields for 'FIRST NAME', 'LAST NAME', 'SSN', 'ID', 'PHONE 1', 'PHONE 2', 'ADDR 1', 'ACCT #', 'TYPE CD', 'TOP STAT', 'VER', 'CAT CD', 'CITY', 'STATE', 'ZIP', and 'ORD #'. There are also checkboxes for 'FT/PT' and 'NEW', and a 'DEL' button. A list of codes is visible on the right side of the form. At the bottom of the form are buttons for 'OKAY', 'APPLY', 'SAVE', 'UNDO', 'HELP', 'DELETE', 'EDIT', 'SELECT', 'BROWSE', and 'ERRORS'. The text 'STUFFTHATHAPPENS.COM BY ERIC BURKE' is at the bottom of the frame.

TYPICAL APPLE PRODUCT...

TOUCH

A GOOGLE PRODUCT...

FIND

YOUR COMPANY'S APP...

FIRST NAME: TYPE CD:
LAST NAME: TOP STAT: ☐
SSN: VER:
ID: FT/PT: ☐ CAT CD:
PHONE 1: CITY:
PHONE 2: STATE:
ADDR 1: ZIP:
ACCT #: ORD #:

4 - K
AA2-
DK9B
KKA?
CN3
AA-9

NEW
DEL

OKAY APPLY SAVE UNDO HELP DELETE EDIT
SELECT BROWSE ERRORS

STUFFTHATHAPPENS.COM BY ERIC BURKE

Planning the software solution

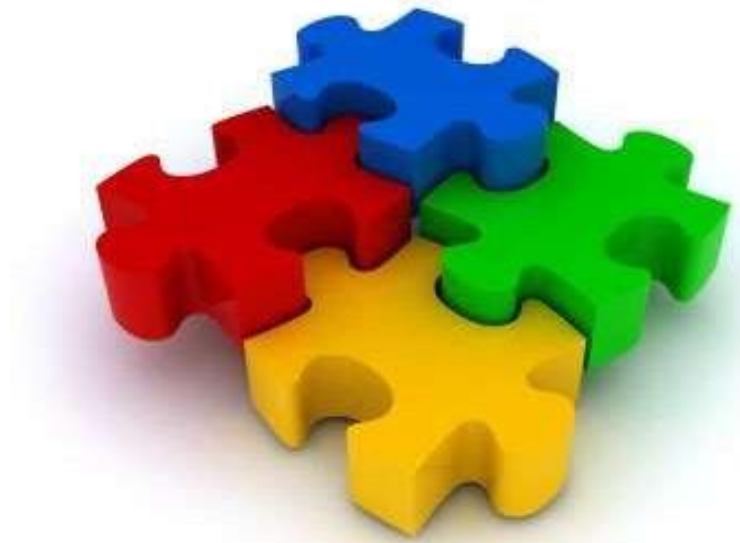
Software Design

Software design is an iterative process through which requirements are translated into a blueprint for constructing the software.

- Initially, the blueprint depicts a holistic view of software.
- During the design process the software specifications are transformed into design models .
- Models describe the details of the data structures, system architecture, interface, and components.

Modularity

- In this concept, software is divided into separately named and addressable components called modules
- Follows “divide and conquer” concept, a complex problem is broken down into several manageable pieces
- Easier to change
- Easier to build
- Easier to maintain

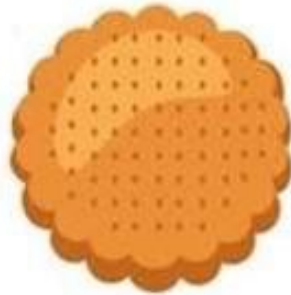


Interface Design

- **UI** is the user interface. This comprises everything a user can see and touch, such as menu options, buttons, text, layouts, navigation elements, sharing options, etc
- **UX** is why you made that change to affect how the user feels and behaves. The user experience is an umbrella term for the user's overall experience with the product: what they liked about it and how easily they accomplished their goals

Interface Design

UI



UX



Interface Design

UI



UX



Open top, shake vigorously



Pray it doesn't get all over you



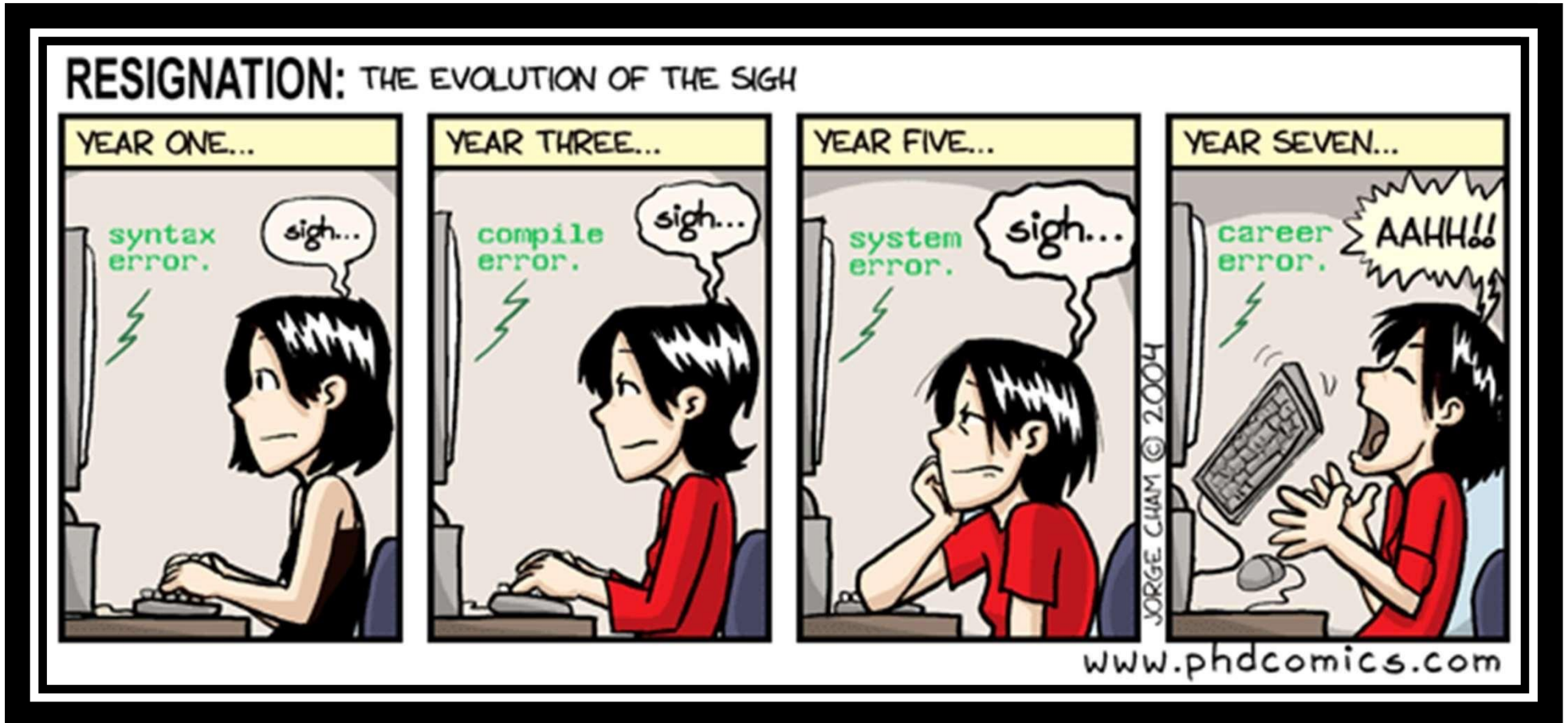
Open top



Squeeze out desired amount



3. Implementation



Code!!!

Refactoring

Refactoring is the process of changing a software system in such a way that it does not alter the external behavior of the code [design] yet improves its internal structure.

When software is refactored, the existing design is examined for

- redundancy
- unused design elements
- inefficient or unnecessary algorithms
- poorly constructed or inappropriate data structures

or any other design failure that can be corrected to yield a better design.



Code Review

- Code review is a software quality assurance process in which software's source code is analyzed manually by a team or by using an automated code review tool.
- The Goal is purely to improving code quality.
- Reviewing the codebase makes sure that every software or new feature developed within the company is of high quality.



4. Testing



Executing the application trying to find software bugs

Verification vs Validation

❖ Verification:

"Are we building the product right".

The software should conform to its specification.

❖ Validation:

"Are we building the right product".

The software should do what the user really requires.



TO BE
CONTINUED...